Jingqi Lu

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Summary

Urban AI engineer with a spatial analytics foundation. I build LLM pipelines and agent systems that make sense of messy city data — architectural archives, cultural event pages, satellite imagery — and turn them into something people can actually use to make decisions.

M.S. Urban Spatial Analytics, UPenn (GPA 3.93), May 2026. Looking for roles where AI meets urban infrastructure, smart cities, or spatial intelligence.

Education

University of Pennsylvania Philadelphia, PA · Aug 2025 – May 2026
M.S. Urban Spatial Analytics (MUSA) · GPA 3.93 / 4.0 (Top 5%) · Graduated May 2026 · Coursework: Statistics & Data Mining · Python Geospatial · Sustainable City AI · Remote Sensing ML · Behavioral & Network Science

University College London London, UK · Sep 2022 – Jun 2025
B.A. Geography (Social Data Science) · 2:1 Honours · Coursework: Data Analysis · Geocomputation · Cartography & Data Viz · Applied ML

University of Amsterdam — Erasmus Exchange Amsterdam, NL · Sep – Dec 2024
Geography & Planning

Experience

Research Assistant · University of Pennsylvania Philadelphia, PA · Feb 2026 – May 2026

- Built a six-agent LangGraph pipeline for automated urban cultural event collection — city parsing, OSM venue discovery, dual-path crawling, and knowledge graph generation via Send API parallel fan-out.
- Designed a hybrid Judge Agent combining rule-based scoring and Claude Sonnet precision evaluation for 0–1 event relevance scoring; adaptive crawl routing escalates to LLM-guided strategy only on low-yield runs, reducing unnecessary API calls.
- Extracted event topic tags and named entities via GPT-4o-mini to build venue-level semantic profiles; mined cross-venue similarity via cosine distance, surfaced through a React + SSE real-time dashboard.

IELTS Teaching Assistant · New Oriental Hefei, China · Jul – Sep 2023

- Led IELTS speaking practice sessions, graded mock exams, and provided written feedback for 20+ students.

Projects

DataTaxonomy — Document Intelligence System · Henning Larsen Philadelphia / Copenhagen · Jan 2026 – Present

- Designed and deployed a four-layer LLM classification pipeline (routing extraction → seven-dimension classification → design impact scoring → risk review) for 3000+ unstructured AEC documents.
- Implemented folder-sibling RAG and project intelligence RAG to recall context from adjacent files and meeting notes, improving project-context awareness in classification outputs.
- Built a four-dimension geometric-mean design impact scoring model (0–100) combining constraint hardness, decision-chain position, phase alignment, and substitutability — driving Review Queue prioritization.
- Designed a dynamic VLM Agent Gate activating vision models only when pre-screen signals detect visual content; integrated multi-LLM/VLM routing via OpenRouter with 10-way parallelism and incremental re-run support.

Co-authored: 'A Data Taxonomy for Integrating Data into Architecture & Planning Decision-Making' (with Nitsan Bartov)

Wildfire Damage Detection — Remote Sensing & Deep Learning · UPenn Philadelphia, PA · Mar 2026

- Trained CNN on EuroSAT benchmark (Sentinel-2 multispectral imagery) for 10-class land use classification; spatial feature learning significantly outperformed traditional ML baselines.
- Built a terrain-aware U-Net semantic segmentation model to predict parcel-level building damage for the 2025 LA Palisades and Eaton wildfires across two geographically distinct fire sites (385K pixels, 5.81% damage rate).
- Constructed 7-channel input tensors (pre/post NIR & SWIR, dNBR, DEM aspect, building footprint), explicitly encoding terrain aspect to distinguish low-sun-angle canyon shadows from true burn signatures.
- Applied block-based spatial K-fold cross-validation to prevent spatial data leakage; Dice + Focal Loss for 16:1 class imbalance — final F1 = 0.653 vs. Random Forest baseline F1 = 0.410.

Skills

AI & LLMs: LangChain · LangGraph · OpenAI API · RAG · Agentic workflows · Hugging Face · OpenRouter · PyTorch · CNN · U-Net · Deep Learning
Data & Geospatial: Python · R · SQL · GeoPandas · ArcGIS Pro · Google Earth Engine · DuckDB · SciPy · Spatial statistics · KMeans/DBSCAN · Time-series
Web & Visualization: JavaScript · React · HTML/CSS · Plotly · Matplotlib · Quarto · Cartographic design
Dev Tools: Git/GitHub · Jupyter · VS Code · SQLite · Google Colab
Languages: English (professional fluency) · Mandarin Chinese (native)